

Algorithm of Linear Regression

Aim: To implement Linear Regression using a salary dataset and calculate the Mean Squared Error (MSE).

Algorithm

1. Start the program.
2. Import the required libraries: Pandas for handling the dataset, Matplotlib for plotting, Linear Regression from *sklearn*, and Mean Squared Error from *sklearn.metrics*.
3. Read the dataset **salary.csv** using Pandas and store it in the DataFrame **df**.
4. Select the input and output variables. **X** represents Years of Experience and **y** represents Salary.

```
X ← Years of Experience  
y ← Salary
```
5. Create a Linear Regression model.
6. Train the model using the input and output data. The model learns the relationship between years of experience and salary.

```
model.fit(X, y)
```
7. Predict the salary values using the trained model.
8. Calculate the Mean Squared Error (MSE) by comparing the actual salary values with the predicted values.

```
y_pred ← model.predict(X)  
MSE ← MeanSquaredError(y, y_pred)
```
9. Display the slope, intercept, and MSE.
10. Plot the original salary data points using a scatter plot and draw the regression line using the predicted values.

```
Plot(X, y)          → Actual data  
Plot(X, y_pred)     → Regression line
```
11. Display the graph with suitable X-axis and Y-axis labels.
12. Stop the program.

Formula Used

Linear Regression: $y = mx + c$

where **y** is the predicted salary, **x** is years of experience, **m** is the slope/coefficient, and **c** is the intercept.

Mean Squared Error: $MSE = (1/n) \sum (y_{\text{actual}} - y_{\text{predicted}})^2$

Here, y_{actual} represents the actual salary and $y_{\text{predicted}}$ represents the predicted salary.